Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 October 2026
Team ID	SKILLWALLET-EV-01
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Ingest and parse 100,000 multi-attribute EV records covering technical, economic, and operational metrics.	High
2	Data Processing	Perform data cleaning, type conversion, duplicate removal, and extract generation in Tableau .hyper format.	High
3	Business Analysis	Compute derived metrics including Cost per Km, Fast Charging Efficiency, and 3-Year Resale Value percentages.	High
4	Business Insights & Analytical Outputs	Generate 9 interactive visualization sheets, a consolidated executive dashboard, and a 4-point narrative story.	High
5	User Interaction & Analysis	Provide global filter controls (Manufacturer, Category, Region, Drivetrain) with real-time responsive updates.	Medium
6	Data Update & Reporting	Embed interactive visualizations into a Flask web application and publish to Tableau Public for public dissemination.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Intuitive, clean UI with clear visual hierarchy, consistent color palettes, and responsive sizing across screen resolutions.
NFR-2	Security	Secure web routing in Flask, sanitization of inputs, and read-only authenticated Tableau Public embedding.
NFR-3	Reliability	100% accurate data representation with zero loss or discrepancy between source CSV and Tableau extracts.
NFR-4	Performance	Dashboard filter actions and visualization rendering completed within 1.5 seconds for 100,000 data

		rows.
NFR-5	Availability	Web application and hosted Tableau Public dashboard ensure 99.9% uptime accessible via standard modern web browsers.
NFR-6	Scalability	System architecture accommodates expanding datasets (up to 1,000,000 records) without architectural overhaul.